

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456954.849415 +/- 0.0001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.002 +/- 0.045
Transit depth (Rp/Rs)^2	0.00043 +/- 0.018 [%]
Orbital Inclination (inc)	80.01 +/- 2.88
Airmass coefficient 1 (a1)	0.5 +/- 0.26
Airmass coefficient 2 (a2)	0.38 +/- 0.41
Scatter in the residuals of the lightcurve fit is	47.41 %
Best Comparison Star	#3 - [223, 162]
Optimal Aperture	1.82
Optimal Annulus	4.75
Transit Duration (day)	0.103 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.002 ± 0.045 vs 0.1178 (deviation 2.57σ)
Duration fit vs published	0.103 d vs 0.125 d (deviation 1.83σ)
Mid-time O-C	-0.2 min
Depth SNR	0.0
Residual scatter	47.41 %

Light curve

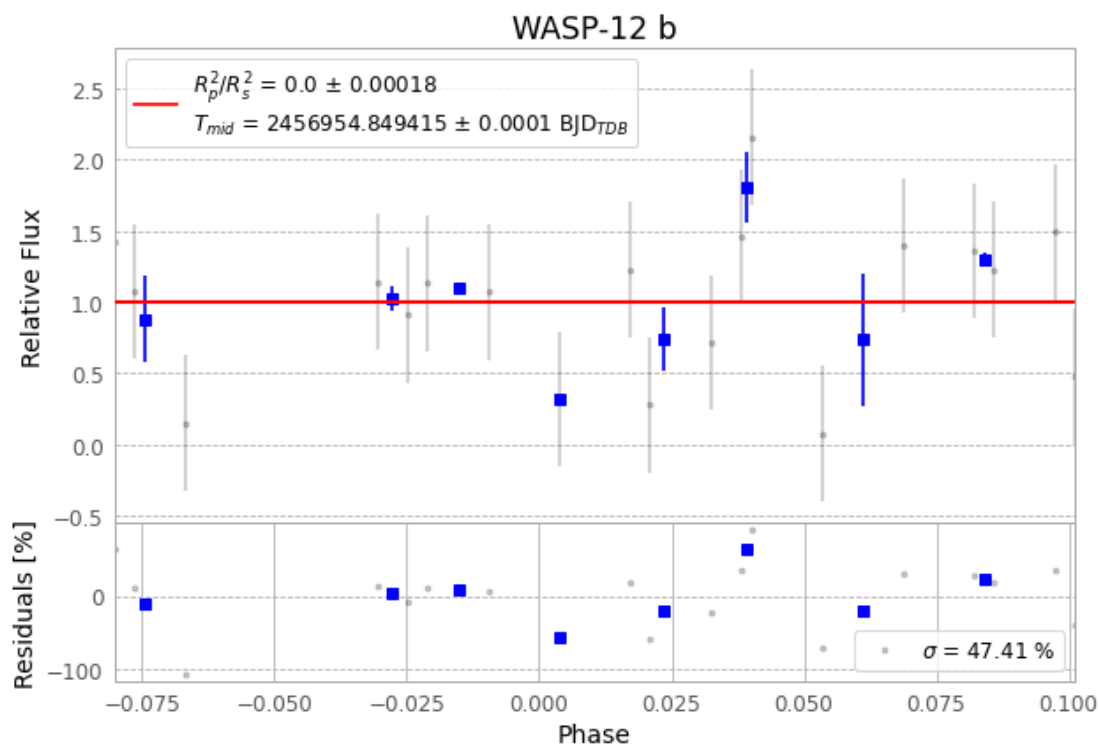


Figure 1 — FinalLightCurve_WASP-12 b_2014-10-23.png (SHA-256 fc422d30b52cfbb1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [188, 316], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 919a1af6b084e46a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

97 FITS frames (64 MB), WASP-12141024061212.FITS → WASP-12141024110012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-141024.log (e9722d3f0bf7...), FinalLightCurve_WASP-12 b_2014-10-23.png (fc422d30b52c...), FinalLightCurve_WASP-12 b_2014-10-23.csv (c78b775dc4a6...)

Compute resources

Wall time 122.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 22:54Z.